



Founders—Mihir Pawar (left), CTO, and Durgesh Garud (right), CEO

wheel conversion kit with a 250W front-wheel motor, a 209Wh Li-ion battery, a compact charger, easy-to-attach pedal-assist sensor, controller box, throttle, wiring harness, and e-brakes. In contrast to the conventional e-cycles, the battery box is not attached to the frame but sits atop the handle, making it suitable for any model cycle. Any cycle can be turned electric in a few steps like swapping the front tyre with the one provided in the kit, attaching the top assembly consisting of battery and controller, attaching the pedal-assist sensor on the crankset, and by fixing e-brakes and throttle. The user can do all this with the help of an instruction manual.

“Since it is a universal kit, one can just use the same kit on dif-

ferent bicycles. It offers a range of around 25km and can charge up to 70% in an hour. The battery box is compact and detachable, enabling the user to charge the battery anytime and anywhere. Therefore, it eliminates range anxiety with the compact charger,” Garud explains.

Speaking about the challenges, Garud says that designing the front-wheel conversion was among the major constraints. “It was not easy to get the motor fixed in the front wheel, especially in India. Since we did not want to source it from other countries, thinking about the cost, we developed it with a manufacturing partner in Pune,” he says.

Garud elaborates, “The placement of the battery is fixed in a comfortable position to give the rider a great handling experience. Design-

ing and developing this was another task, which took a lot of research and development. Finally, the battery has to be charged up to 70% in an hour, which was also customised, developed, and is being sourced from a partner in India itself.”

While Garud appreciates the government’s Faster Adoption and Manufacturing of Hybrid & Electric Vehicles (FAME) scheme, he also points out the terms that limit the scheme to those vehicles offering a minimum range of 80km and a minimum speed of 40kmph. He suggests that including low-speed micro-mobility vehicles in this scheme will be a great boost to the market and mass adoption of electric vehicles.

As a startup founder, Garud says patience and perseverance are his best lessons in the journey. He advises the others, “Focus on sustainable growth. Don’t compromise on building a great foundation for your startup just for short-term financial gratification. It is really important to get the foundation correct and make sure the product or service you offer adds value to your consumers.”

Currently, the firm is operating on direct to consumer (D2C) model, allowing customisation with customer requirements for initial orders. The firm is planning to conduct the next round of funding and is also open to welcome like-minded partnerships. “We have plans to get launched on various e-commerce platforms and in a few retail stores. We also have bigger goals to step into the UAE market soon,” he says.

3 VFLYX DRONE AT YOUR SERVICE

With India focusing entry in drone manufacturing with its own drone portals, regulations, and licenses, VFLYX—a Bangalore based startup—is already into manufacturing customised UAVs/drones using in-house hardware and on-board flight software

The parent firm, Xboom, started in 2015 with focus on addressing safety and security products. It eventually started exploring the next level into surveillance using drones. Introducing VFLYX in 2021, the firm has now ventured exclusively into drone sales and operations to cater